🌡️ Hypothalamus & Pituitary: Cram Sheet NSC 4351 · if I see X, I answer Y

🎯 He said know this

- Two control routes: **neural** (fast) + **hormones in blood** (slow, long lasting)
- **Memorize only 2 hormones:** posterior pituitary **oxytocin + vasopressin (ADH)**. "Not a physiology class"; anterior chart = FYI
- **Front turns things down, back turns things up**
- **Most input AND most output = brainstem.** Cortex input is minor
- Pituitary tumor patients complain of **vision**, not hormones

🧬 Pituitary: two lobes

	Anterior (adenohypophysis)	Posterior (neurohypophysis)
Origin	**Rathke's pouch** (roof of mouth)	Floor of diencephalon
Brain?	**NO**	**YES** (hypothalamic axons)
Link	**Blood:** median eminence → **hypophyseal portal veins** → anterior lobe ("train + transfer bus")	**Nerves:** SO + PV axons down the stalk
Hormones	PRL, FSH/LH, TSH, ACTH, GH (FYI)	**Oxytocin, ADH**

TRAP: oxytocin + ADH are **made** in **supraoptic + paraventricular** nuclei. Posterior pituitary only **stores + releases**.
Portal = capillary bed → veins → capillary bed (short route, skips the heart).

💧 The 2 hormones

Vasopressin (ADH)	kidney holds water; squeezes vessels → **blood pressure**
Oxytocin	**milk letdown**, uterine contractions, **bonding**
No ADH	**Central diabetes insipidus**: lots of dilute urine, thirst, **↑ serum sodium** (the clinical test). Not sugar diabetes

🔀 3 divisions (front → back)

Division	Nuclei	Does
A Preoptic (front)	**Supraoptic, suprachiasmatic, paraventricular**	ADH/oxytocin, clock, heat loss, sleep switch
B Tuberal (mid)	**Ventromedial, dorsomedial, arcuate** + median eminence	Satiety / weight set point, releasing factors
C Mammillary (back)	**Posterior nucleus** + mammillary nuclei	Heat conservation, sympathetic

- "Pre-op" = before the optic chiasm = front
- **White round thing in hypothalamus = fornix** (axons). Landmark: splits **medial vs lateral** hypothalamus

🔌 Inputs / Outputs

From **orbitofrontal cortex** (emotional PFC)	**MFB** (medial forebrain bundle)
From **brainstem** (main input)	**DLF** + MFB
To **brainstem + spinal cord** (MOST output)	**DLF**
To anterior thalamus	**Mammillothalamic tract**
Amygdala ↔	Stria terminalis (VAP dotted line = **ignore**)
Hippocampus →	Fornix → mammillary bodies

🚪 Circumventricular organs (no BBB, leaky capillaries)

OVLT + **subfornical organ**	sense low volume / salt → SO + PV → **ADH**
Area postrema (4th ventricle)	toxin in blood → **vomit**
Median eminence · neurohypophysis · pineal	releasing factors · ADH/oxytocin out · melatonin

Low blood volume: kidney **renin** → OVLT + SFO → SO/PV → ADH

⚙️ Functions

Anterior hypothalamus	→ brainstem **parasympathetic** (turn down)
Posterior hypothalamus	→ lateral horn, **intermediolateral cell column**: **sympathetic** (turn up)
Circadian	retina → **SCN** (sits on the chiasm) → **pineal**
Endocrine	releasing factors → portal system then anterior pituitary
Lateral n.	**induces** feeding · lesion → stops eating
Ventromedial n.	**inhibits** feeding (satiety) · lesion → obese
Paraventricular n.	**integrates**
Arcuate	reads leptin/insulin: **NPY/AgRP** eat ↑, **POMC** eat ↓

🌡️ The AC vs the Heater

	❄️ Anterior (preoptic) = AC	🔥 Posterior = Heater
Job	heat **loss**	heat **conservation**
Stim	dilate skin vessels (red), sweat, pant, stop shivering	shiver, constrict skin vessels (pale), thyroid ↑
Lesion	**Hyperthermia**	**Hypothermia** (if in cold)

- Both run **at the same time** (fast temp changes)
- **Heat stroke = AC fails, heater still on** → temp keeps rising despite cooling (103 → 105.7). Emergency: brain damage, death
- **Heat exhaustion** = dizziness + headache → stop, cool place, sit, cool off
- Older people: AC breaks with age → more at risk

🩺 If I see X → I answer Y

Can't see out to the sides, can't count fingers laterally	**Pituitary tumor on chiasm** (bitemporal)
Pregnant, sudden sharp headache behind eyes	**Pituitary apoplexy** (stroke) · **anterior** (A)
Kid giggles, stares, freezes	**Gelastic seizure** → **hypothalamic hamartoma**
Cyst at sella from mouth-roof tissue	**Rathke cleft cyst** → **marsupialize** (open wide to drain)
Huge dilute urine + high sodium	**Central DI** (no ADH)
Wasting infant, hypothalamic mass	**Diencephalic syndrome** → neuro early, tube-feed
Big jaw, hands, brow (adult)	**Acromegaly** (GH ↑ after plates close) · before = gigantism
Moon face, striae, truncal obesity	**Cushing disease** (ACTH/cortisol ↑)
Look-alike kids, goiter	**Hypothyroidism** (iodine-poor area)
Temp rising despite ice + IV	**Heat stroke**
Front lesion / back lesion	**too hot** / **too cold**